Data Encryption Standard

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DES Block Cipher

The Data Encryption Standard (DES):

This algorithm adopted in 1977 by the National Institute of Standards and Technology (NIST). The algorithm itself is referred to as the Data Encryption Algorithm (DEA). For DES, data are encrypted in 64-bit blocks using a 56-bit key. The algorithm transforms 64-bit input in a series of steps into a 64-bit output. The same steps, with the same key, are used to reverse the encryption.

DES encryption algorithm:

The general structure of the DES consists of (1) key schedule, (2) round function and (3) initial and final permutation.

Step1: **Plaintext is broken into blocks of length 64 bits.**

Step2: **The 64-bit block undergoes an initial permutation (IP) using initial permutation IP table, IP(M).**

Step3: **The 64-bit permuted input is divided into two 32-bit blocks: left (L) and right (R). The initial values of the left and right blocks are denoted L_0 and R_0.**

Step4: **There are 16 rounds of operations on the L and R blocks. During each round, the following formula is applied:**

$L_n = R_{n-1}$

$R_n = L_{n-1}$ XOR $F(R_{n-1}, K_n)$

DES Block Cipher

Step5: The function **F(.)** represents the heart of the DES algorithm. This function implements the following operations:

1-Expansion: The right **32-bit** half-block is expanded to **48 bits** using the **expansion permutation** (**E**) table, $E(R_{n-1})$.

2-Key mixing: The expanded result is combined with a **subkey** using an XOR operation. Sixteen 48-bit subkeys (one for each round) are derived from the main key using the **key schedule**, $K_n + E(R_{n-1})$.

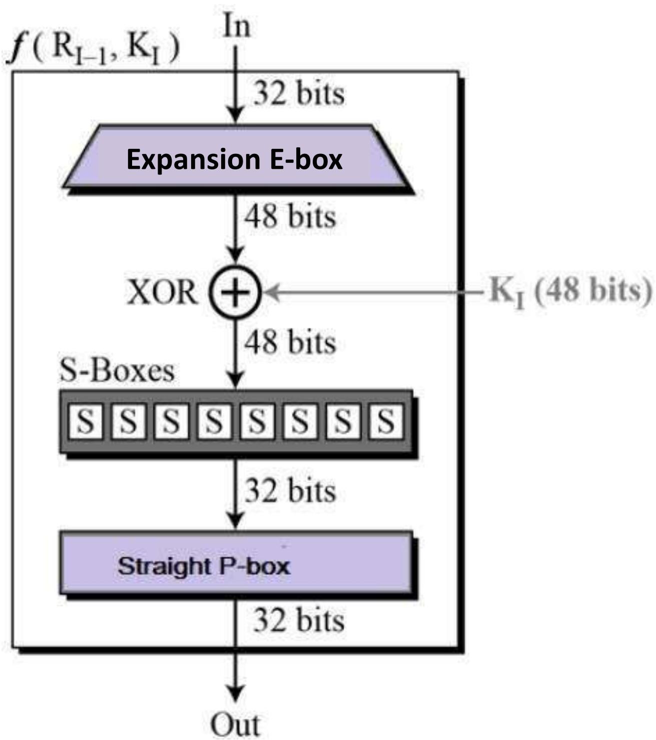
3-Substitution: After mixing in the subkeys, the block is divided into eight 6-bit pieces and fed into the substitution boxes (S-boxes), which implements nonlinear transformation. Each 6-bit piece uses as an address in the **S-boxes** where the first and last bits are used to address the i^{th} row and the middle four bits to address the j^{th} column in the S-boxes. The output of each **S-box is 4-bit length** piece. The output of all **eight S-boxes** is then combined into **32** bit section.

$K_n + E(R_{n-1}) = B_1B_2B_3B_4B_5B_6B_7B_8$

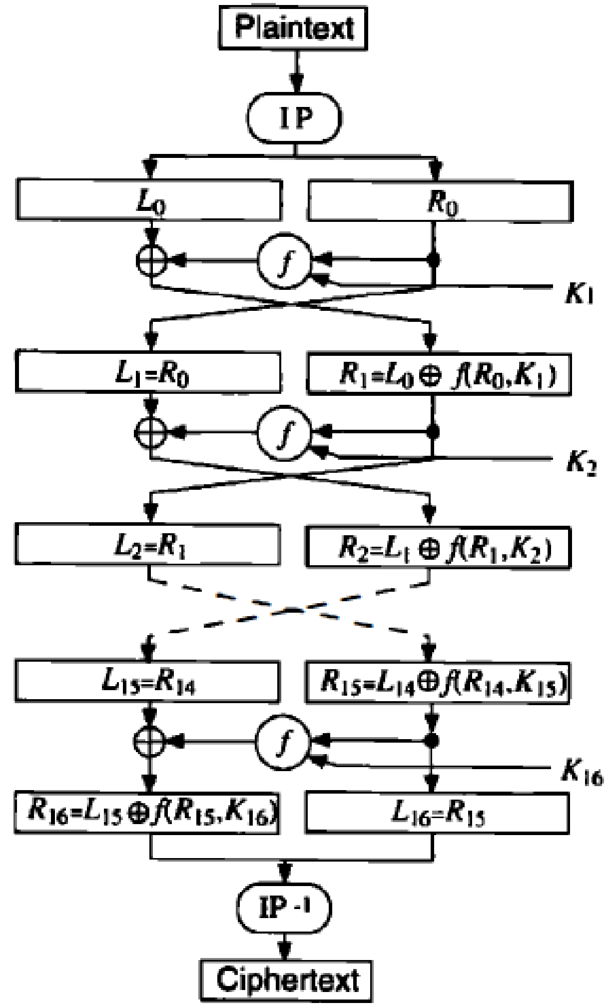
$S(Kn + E(Rn-1)) = S1(B_1)S2(B_2)S3(B_3)S4(B_4)S5(B_5)S6(B_6)S7(B_7)S8(B_8)$

4-Permutation: The **32** bits outputs from the S-boxes are rearranged using the **P-box**, $F=P(S(K_n + E(R_{n-1})))$

Step6: The results from the final DES round (i.e., L_{16} and R_{16}) are recombined into a 64-bit value and rearranged using an inverse initial permutation **(IP^{-1})** table. The output from IP^{-1} is the 64-bit ciphertext block.



Single Round function (F) of the DES



DES Encryption Flowchart

(a) Initial Permutation (IP)

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

(c) Expansion Permutation (E)

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

(b) Inverse Initial Permutation (IP^{-1})

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

(d) Permutation Function (P)

16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

Tables used in the DES algorithm

DES Block Cipher

Key schedule (generator):

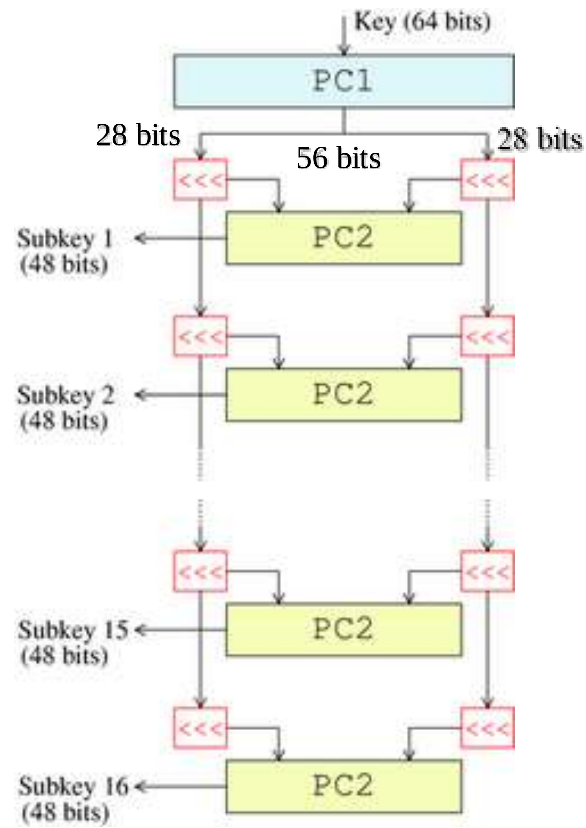
This algorithm generates the subkeys ($K \rightarrow K_1, K_2 \ldots K_{16}$).

1- The 56 bits of the key are selected from the initial 64 by Permuted Choice 1 (PC1) table.

2- The 56 bits are divided into two 28-bit halves.

3- In each round, both halves are rotated left by one or two bits (specified for each round).

4- The 48 subkey bits are selected by Permuted Choice 2 (PC2) table (24 bits from the left half, and 24 from the right) and used in each round.



Key schedule structure

General remarks in the DES:

1- The S-boxes provide the core of the security of DES and the cipher would be linear, and trivially breakable without them.

2- The substitution and permutation in the DES provide confusion and diffusion.

The DES S-Boxes

S_1	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13
S_2	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9
S_3	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12
S_4	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14
S_5	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3
S_6	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13
S_7	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12
S_8	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

(a) Input Key

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

(b) Permuted Choice One (PC-1)

57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22
14	6	61	53	45	37	29
21	13	5	28	20	12	4

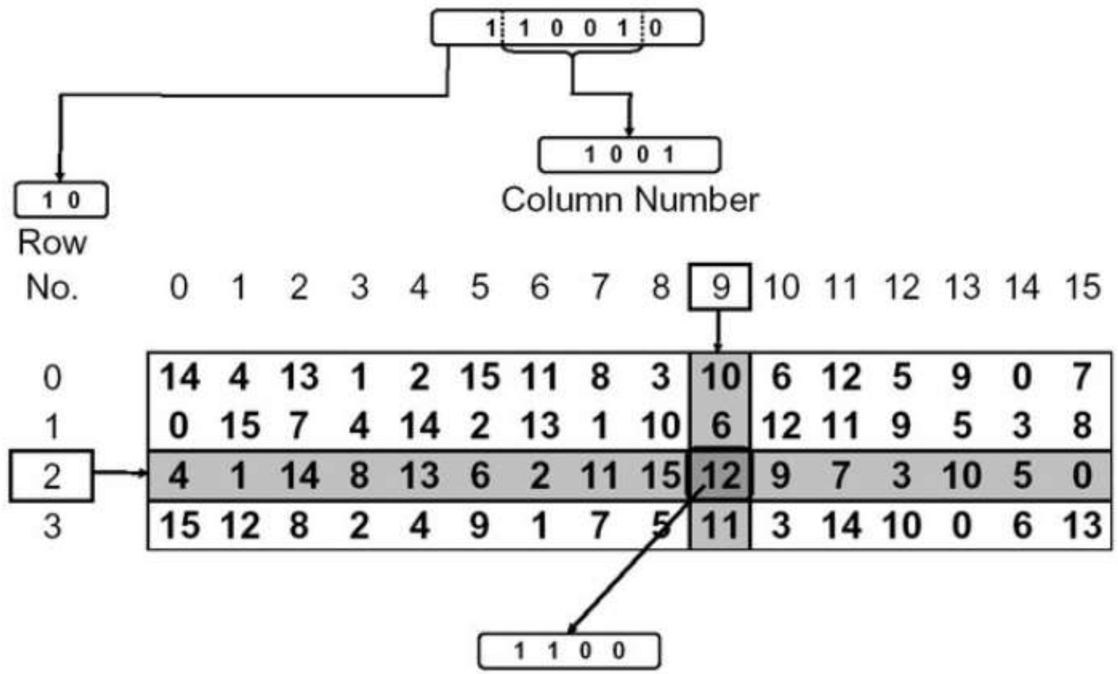
(c) Permuted Choice Two (PC-2)

14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

(d) Schedule of Left Shifts

Round Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bits Rotated	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

Tables used in DES key generator



Application of S-box in DES Algorithm

DES Block Cipher

DES decryption :

The decryption algorithm uses the same steps exactly as in the encryption algorithm except that the application of the subkeys is reversed (i.e. in round1 use K_{16}, round2 use K_{15} and so on).

Security and cryptanalysis:

The two most widely used attacks on block ciphers are linear and differential cryptanalysis. DES is also vulnerable to a brute-force (exhaustive search) attack.

Triple DES:

In cryptography, Triple DES (3DES or TDES), officially the Triple Data Encryption Algorithm (TDEA or Triple DEA), is a symmetric-key block cipher, which applies the DES cipher algorithm three times to each data block.

Therefore, Triple DES uses a "key bundle" that comprises three DES keys, $K1$, $K2$ and $K3$, each of 56 bits.
The encryption algorithm is:

$$\text{ciphertext} = E_{K3}(D_{K2}(E_{K1}(\text{plaintext}))).$$

That is, DES encrypt with $K1$, DES *decrypt* with $K2$, then DES encrypt with $K3$.
Decryption is the reverse:

$$\text{plaintext} = D_{K1}(E_{K2}(D_{K3}(\text{ciphertext}))).$$

That is, decrypt with $K3$, *encrypt* with $K2$, then decrypt with $K1$.
Each triple encryption encrypts one block of 64 bits of data.